

Module 3 Simulation: Lunar Greenhouse

What this simulation does

The Lunar Greenhouse puts you in charge of a crop through a full growth cycle on the Moon. You manage four environmental variables across four growth phases: Germination, Vegetative Growth, Flowering, and Harvest. At each phase transition, you adjust your settings before advancing. Get the conditions right and the crop advances. Get them wrong and the crop degrades or dies.

The simulation enters fullscreen mode automatically when you confirm your crop selection. This is intentional — it creates a focused environment that mirrors the isolation and stakes of real mission operations.

Selecting your crop

When the simulation loads, a crop selection modal appears. You will see a grid of available crops. Currently active crops include Corn, Pepper, Tomato, and Zucchini. Click on a crop to see its profile before committing.

Each crop profile shows:

Harvest time: how many days from planting to harvest.

Mature size: how large the plant gets.

Difficulty rating: how precisely the variables need to be managed.

Growing requirements: what the crop specifically needs for light, water, and nutrients.

Space relevance: why this crop matters for long-duration missions.

Click START GROWING to confirm your selection and begin.

The four growth phases

Phase 1 — Germination: Seeds need warmth and consistent moisture to break dormancy and establish roots. This phase is typically the most sensitive to temperature. Light requirements are lower than later phases.

Phase 2 — Vegetative Growth: The plant is building its structure — stems, leaves, root system. Strong light and consistent nutrition drive healthy development. Weak light at this stage causes leggy, fragile plants that underperform in later phases.

Phase 3 — Flowering: This is the highest-stakes phase for fruiting crops. Flowers open and early fruit begins to set. Water stress during flowering causes blossom drop and poor fruit set. This is the most critical water period for most crops in the simulation.

Phase 4 — Harvest: The crop reaches maturity. Conditions in this phase affect final yield quality. Maintaining stable conditions through to harvest determines whether your mission gets a full

nutritional return from the crop.

The four variables you control

Light: LOW, MED, or HIGH. Controls grow light intensity. Most crops need higher light during Vegetative and Flowering phases than during Germination.

Water: LOW, MED, or HIGH. Controls irrigation frequency. Fruiting crops typically need their highest water input during Flowering.

Nutrients: LOW, MED, or HIGH. Controls fertilizer concentration in the water supply. Excess nutrients cause burn. Deficient nutrients stunt growth.

Temperature is also tracked. The simulation monitors whether your settings maintain the crop's preferred temperature range. Each crop has a different optimal temperature window shown in its profile.

Reading the health meter

A health bar on the right panel shows your crop's current status. The bar has four labeled zones: Critical, Stressed, Healthy, and Thriving. Both the color of the bar and a text label update as conditions change. You do not need to read color alone — the text label always tells you where you stand.

If the health bar drops to Critical and you do not adjust your settings, the crop will die and the scenario will end. You can always restart with the same crop or try a different one.

How to advance between phases

Set your three variables using the buttons in the left control panel. When you are satisfied with your settings, click START PHASE to run the current phase. The simulation runs the phase and updates the health bar based on how well your settings matched the crop's needs.

A phase result message appears telling you what happened and why. Read it carefully before adjusting for the next phase. What worked in Germination may not be correct for Vegetative Growth — requirements shift between phases.

Adjust your settings for the next phase and click START PHASE again to continue.

Connecting this to your assignment

Your Mission Operations Report asks for a growth cycle overview, a description of your critical phase transitions (where did health change most dramatically and why), what worked and what did not, and recommendations for the next crew. Take notes as you play — what settings you used each phase and what the outcome was. That information becomes your report.

The most useful runs are the ones where something goes wrong. A crop that struggles in Flowering and then partially recovers in Harvest gives you more to write about than a perfect run.

If you get stuck

Go back to the crop profile and re-read the growing requirements. Each crop tells you what it needs. If your crop is in Critical health, change the variable that deviates most from the profile's recommended range. You cannot break the simulation — restart as many times as you need. The simulation does not save progress between sessions, so if you close the browser window, you will start fresh from crop selection.